Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2023 | ENGI3451-WE01 |

**Title: (BEng/MEng)**

Electronics and Communications 3

| | |
|---|---|
| Time Allowed: | 2 hours |
| Additional Material provided: | No |
| Materials Permitted: | N/A |
| Calculators Permitted: | Those from the Casio fx-83 and fx-85 series. |

| | |
|---|---|
| **Instructions to Candidates:** | This is an in-person, invigilated exam.<br><br>Answer ALL questions.<br>Use a separate answer book for each question.<br>All relevant workings must be shown. |

| Revision: | |
|---|---|

# Section A

## Question 1

(a) Consider that a message signal $m(t) = 1.5\cos(8 \times 10^3 \pi t)$ is frequency modulated (FM) with a frequency sensitivity factor $k_f = 200$ Hz/V. Assume that the indirect method is used to generate the wideband (WB) FM signal. The carrier frequency used to generate the narrowband (NB) FM signal is $f_{c,NB} = 100$ kHz. For the WB FM signal, the desired carrier frequency is $f_c = 90$ MHz and the desired frequency deviation is $\Delta f = 60$ kHz.

(i) Determine the frequency deviation of the NB FM signal.

(ii) Determine the multiplying factor for the frequency multiplier.

(iii) Determine the frequency of the local oscillator of the mixer.

(iv) Determine the effective bandwidth of the WB FM signal.

**[50%]**

(b) Consider an amplitude modulation (AM) broadcasting system where the AM signal of interest at the input of a superheterodyne receiver has a carrier frequency of $f_c = 1400$ kHz. The intermediate frequency is $f_{IF} = 455$ kHz.

(i) Determine the frequency of the local oscilator so that the superheterodyne receiver can recover the message signal from the AM signal of interest.

(ii) Determine the frequency of the image signal that can also be translted to the intermediate frequency $f_{IF} = 455$ kHz.

(iii) Explain how the image signal can be removed.

**[30%]**

(c) Consider a communication system with a bandwidth of 20 MHz and a white noise with a spectral density of $10^{-7}$ W/Hz. Determine the required signal power so that a channel capacity of 10 Mbps is obtained.

**[20%]**

**continued**

## Question 2

(a) Consider a sinusoidal carrier signal with a frequency $f_c = 1.2$ MHz and a message signal $m(t) = 1.5\cos(3 \times 10^3 \pi t) + 2\sin(1.5 \times 10^3 \pi t + \frac{\pi}{2})$. Assume double-sideband amplitude modulation (DSB-AM).

(i) Explain how to achieve a desired modulation index of $0.9$.

(ii) Determine the bandwidth of the modulated signal.

(iii) If an envelope detector is used to recover the message signal, determine the optimum choice of the RC time constant.

**[35%]**

(b) A discrete memoryless source produces outputs $s_1$, $s_2$, $s_3$, $s_4$, $s_5$, $s_6$, $s_7$ and $s_8$ with probabilities $0.16$, $0.125$, $0.3$, $0.125$, $0.0975$, $0.0675$, $0.0625$ and $0.0625$, respectively. Assume coding over an alphabet of size $D = 3$ (ternary). Note that a ternary code is one which uses three alphabets in coding, e.g., $\{0, 1, 2\}$.

(i) Design a Huffman code for this source.

(ii) Determine the average codeword length.

(iii) Determine the entropy of this source. Comment on the result.

**[50%]**

(c) Consider a (7,3) maximum-length code. Determine the generator matrix of this code in the systematic form.

**[15%]**

**continued**

# Section B

## Question 3

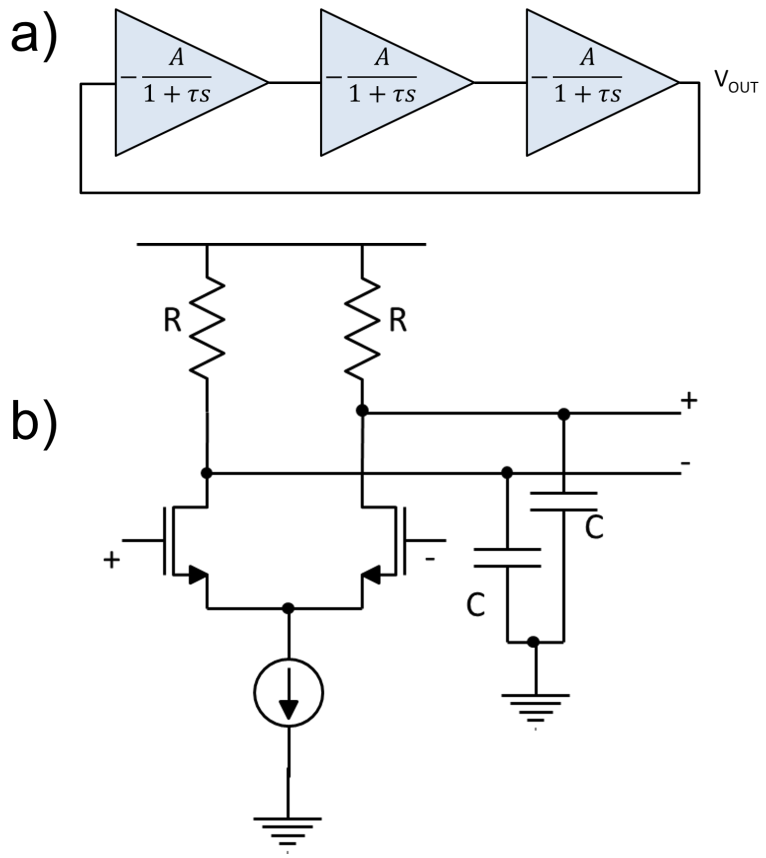



Figure Q3: Ring oscillator diagram a) and circuit used for each stage b).

Figure Q3-a) shows the diagram of a 3-stage ring oscillator, where $A$ is the DC gain and $\tau$ the dominant time constant of each stage.

(a) Using the Barkhausen criterion, determine an expression for the oscillator frequency and minimum gain for each stage. Clearly show all your working. **[40%]**

(b) The circuit used for the three stages is shown in Figure Q3-b). Assume the following values and MOSFETs' parameters:

- $\mu_n C_{OX} = 120\ \mu\text{A/V}^2$
- $V_T = 0.5$ V
- $V_{DD} = 3$ V
- $I_Q = 100\ \mu$A

**continued**

- $W/L = 10$

You can neglect channel-modulation effects and any parasitic elements introduced by the MOSFETs (including the gate capacitance).

(i) Find the highest gain and the corresponding value of $R$ that this circuit can achieve. Assume that the gate voltage of both MOSFETs is $V_{DD}/2$. Hint: remember that if the voltage on the drain is below a certain value the MOSFET is no longer in saturation.

(ii) Choose appropriate values for $R$ and $C$ to achieve an oscillation frequency $f_0 = 100$ MHz.

**[40%]**

(c) Propose a change in the circuit of this oscillator to enable the control of the frequency using an external voltage.

**[20%]**

## Question 4

(a) Explain, in your own words, how a CMOS inverter operates, with clear reference to its output characteristic (i.e. $V_{OUT}$ vs. $V_{IN}$) and its main advantages when compared to a simple common-source amplifier.

**[20%]**

(b) We want to design a parity checker operating on a 3-bit input (ABC). If the number of input "1" is even (or zero) the output (OUT) is "1". In all other cases the output is "0". The truth table is:

| A | B | C | OUT |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

(i) Draw a CMOS circuit which implements this combinatorial logic circuit.

(ii) Assume that a prototype inverter has already been designed; the N-MOSFET has an aspect ratio $(W_N/L_N)$ = (64 nm/32 nm) and the P-MOSFET has an aspect ratio $(W_P/L_P)$ = (256 nm/32 nm). Dimension all MOSFETs in the combinatory logic circuits in order to achieve a symmetric operation and a driving capacity as the prototype inverter in the worst case.

**[60%]**

(c) Assume that an integrated circuit consists of a million CMOS inverters, with their input connected to the same clock signal at $f_{CLK} = 1$ GHz. Assuming the supply voltage $V_{DD} = 1$ V and that each inverter drives a capacitive load $C_L = 1$ fF (i.e. $10^{-15}$ F), find the total dynamic power dissipated by the circuit.

**[20%]**

**END**